

Name: \_\_\_\_\_

# NUMBER OPERATIONS, (BIDMAS / BODMAS)

Date: \_\_\_\_\_

**LO: I can apply the correct order of operations (BIDMAS) to evaluate expressions.**

## Order of operations

**BIDMAS** tells us the order: **B**rackets, **I**ndices, **D**ivision/  
**M**ultiplication, **A**ddition/**S**ubtraction.

**Key point:** Division and multiplication are equal priority (left to right). Addition and subtraction are equal priority (left to right).

## BIDMAS in action

|   |                         |                             |
|---|-------------------------|-----------------------------|
| ● | $3 + 4 \times 2 = 11$   | multiply first              |
| ● | $(3 + 4) \times 2 = 14$ | brackets first              |
| ● | $2^2 + 3 = 7$           | indices before addition     |
| ● | $3 + 4 \times 2 = 14$   | left-to-right is wrong here |
| ● | $8 \div 2(2+2) = 1$     | common viral debate         |

## Key idea

Always follow BIDMAS: Brackets first, then Indices (powers), then Division/Multiplication (L to R), then Addition/Subtraction (L to R).



**Brackets, Indices, Divide/Multiply, Add/Subtract**

## MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

### Example 1 - multiplication before addition

Calculate  $5 + 3 \times 4$

Multiplication first:  $3 \times 4 = 12$

Then add:  $5 + 12$   
 $= 17$

**= 17**

Without brackets,  $\times$  and  $\div$  always come before  $+$  and  $-$ .

### Example 2 - brackets first

Calculate  $(8 + 2) \times 3$

Brackets first:  $8 + 2 = 10$

Then multiply:  $10 \times 3$   
 $= 30$

**= 30**

Brackets override the normal priority.

### Example 3 - indices

Calculate  $4 + 3^2 \times 2$

Indices first:  $3^2 = 9$

Multiply:  $9 \times 2 = 18$

Add:  $4 + 18 = 22$

**= 22**

Indices before multiplication, multiplication before addition.

### Example 4 - complex expression

Calculate  $20 - 2 \times (3 + 4)^2 \div 7$

Brackets:  $3 + 4 = 7$

Indices:  $7^2 = 49$

Left to right:  $2 \times 49 = 98$ ;  $98 \div 7 = 14$ ;  
 $20 - 14 = 6$

**= 6**

Work through BIDMAS step by step.

## YOUR TURN – PRACTICE (deliberate practice)

### 1. Multiplication and addition

Calculate using BIDMAS.

- a)  $2 + 5 \times 3$
- b)  $10 - 4 \times 2$
- c)  $6 \times 2 + 4$
- d)  $20 - 3 \times 5$

### 2. Brackets

Do brackets first.

- a)  $(4 + 3) \times 2$
- b)  $5 \times (8 - 3)$
- c)  $(10 + 5) \div 3$
- d)  $(6 - 2) \times (3 + 1)$

### 3. With indices

Remember: indices come after brackets but before  $\times$  and  $\div$ .

- a)  $2^2 + 3^2$
- b)  $5 \times 2^2$
- c)  $(2 + 3)^2$
- d)  $10 - 3^2$

### 4. Multi-step

Evaluate step by step.

- a)  $3 + 4 \times 2 - 1$
- b)  $12 \div 4 + 2 \times 5$
- c)  $(5 + 3)^2 \div 4$
- d)  $2 \times (10 - 4)^2 \div 3$

### 5. Mixed BIDMAS

Show your working clearly.

a)  $8 + 12 \div (2 + 4)$

b)  $3^2 \times 2 + 4$

c)  $100 - (4 + 6)^2$

d)  $5 \times 3 + 2^3$

e)  $(7 - 3)^2 + (2 + 1)^2$

## COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a)  $3 + 4 \times 2 = 11$  ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

b)  $3 + 4 \times 2 = 14$  ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

c) Brackets are always done first

☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

d) Addition comes before subtraction ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

e) Indices are done before multiplication ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

## 6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Insert brackets to make each statement true: (i)  $3 + 4 \times 5 = 35$ , (ii)  $12 \div 2 + 4 = 2$ , (iii)  $2 + 3^2 = 25$ .

Expression: \_\_\_\_\_

Simplified: \_\_\_\_\_

b) Using the digits **2, 3, 4, 5** (each exactly once) and the operations  $+$ ,  $-$ ,  $\times$ ,  $\div$ , write an expression that equals **1**. You may use brackets.

Expression: \_\_\_\_\_

Simplified: \_\_\_\_\_

### TOP TIPS

Read the question carefully. Show all your working step by step.  
Check your answer makes sense.

### CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

